

Lecture 17: State Classification, cont.

APRIL 8

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17.1 State Classification

Definition 17.1. We say x communicates with y and write $x \rightarrow y$ if there is a positive probability of reaching y from x :

$$\rho_{xy} = \mathbb{P}_x(T_y < \infty) > 0.$$

Theorem 17.2. If $x \rightarrow y$ and $y \rightarrow z$, then $x \rightarrow z$.

Theorem 17.3. If $\rho_{xy} > 0$, but $\rho_{yx} < 1$, then x is transient.

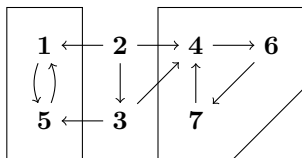
Remark 17.4. Stated a different way, if I can get from x to y , but I am not guaranteed to get from y to x , then I could get lost and never return to x .

Theorem 17.5. If x is recurrent and $\rho_{xy} > 0$, then $\rho_{yx} = 1$.

Remark 17.6. This is the contrapositive of the previous. But also you can think of it as "If I know I return to x infinitely many times, and I can get from x to y , then I have to get to from y back to x ."

Example 17.7.

	1	2	3	4	5	6	7
1	.7	0	0	0	.3	0	0
2	.1	.2	.3	.4	0	0	0
3	0	0	.5	.3	.2	0	0
4	0	0	0	.5	0	.5	0
5	.6	0	0	0	.4	0	0
6	0	0	0	0	0	.2	.8
7	0	0	0	1	0	0	0



The state 2 communicates with 1, which does not communicate with it, so Theorem 1.5 implies that 2 is transient. Likewise 3 communicates with 4, which doesn't communicate with it, so 3 is transient. To conclude that all the remaining states are recurrent we will introduce two definitions and a fact.

Definition 17.8. A set A is closed if it is impossible to get out, i.e., if $i \in A$ and $j \notin A$ then $p(i, j) = 0$.

Remark 17.9. In Example 1.14, $\{1, 5\}$ and $\{4, 6, 7\}$ are closed sets. Their union, $\{1, 4, 5, 6, 7\}$ is also closed. One can add 3 to get another closed set $\{1, 3, 4, 5, 6, 7\}$. Finally, the whole state space $\{1, 2, 3, 4, 5, 6, 7\}$ is always a closed set.

Definition 17.10. Among the closed sets in the last example, some are obviously too big. To rule them out, we need a definition. A set B is called *irreducible* if whenever $i, j \in B$, i communicates with j .

Remark 17.11. The irreducible closed sets in the Example 1.14 are $\{1, 5\}$ and $\{4, 6, 7\}$. The next result explains our interest in irreducible closed sets.

Theorem 17.12. If x is recurrent and $x \rightarrow y$, then y is recurrent.

Theorem 17.13. In a finite closed set there has to be at least one recurrent state.

17.2 Stationary Distributions

In the next section we will see that if we impose an additional assumption called aperiodicity an irreducible finite state Markov chain converges to a stationary distribution

$$p^n(x, y) \rightarrow \pi(y).$$

Question 1. What happens in a Markov chain when the initial state is random?

Breaking things down according to the value of the initial state and using the definition of conditional probability

$$\begin{aligned} \mathbb{P}(X_n = j) &= \sum_i \mathbb{P}(X_0 = i, X_n = j) \\ &= \sum_i \mathbb{P}(X_0 = i) \mathbb{P}(X_n = j | X_0 = i) \end{aligned}$$

If we let $q(i) = \mathbb{P}(X_0 = i)$, then we get

$$\mathbb{P}(X_n = j) = \sum_i q(i) p^n(i, j).$$

A stationary distribution is a distribution $\pi = (\pi_1, \pi_2, \dots, \pi_n)$ (where $\pi_1 + \pi_2 + \dots + \pi_n = 1$, after all it is a distribution) such that taking one more step through the transition matrix p , we end up with the same distribution as before, namely π . This can be represented by $\pi p = \pi$. We will explain this concept through 2 and 3 state Markov chains, and then generalize to a method that works for any size Markov Chain.

Example 17.14. Weather in Ithaca, NY Let X_n be the weather on day n in Ithaca, NY, which we assume is either: 1 = rainy, or 2 = sunny. Even though the weather is not exactly a Markov chain, we can propose a Markov chain model for the weather by writing down a transition probability

	1	2
1	0.6	0.4
2	0.2	0.8

The table says, for example, the probability a rainy day (state 1) is followed by a sunny day (state 2) is $p(1, 2) = 0.4$. A typical question of interest is

Question 2. What is the long-run fraction of days that are sunny?

To answer that we ask ourselves that the initial distribution is $q(1) = 0.3$ and $q(2) = 0.7$, meaning 30% of the time it is rainy, and 70% of the time its sunny. In this case

$$(0.3 \quad 0.7) \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix} = (0.32 \quad 0.68)$$

since $0.3(0.6) + 0.7(0.2) = 0.32$
 $0.3(0.4) + 0.7(0.8) = 0.68$

Example 17.15. Consider the social mobility chain and suppose that the initial distribution: $q(1) = 0.5, q(2) = 0.2$, and $q(3) = 0.3$. Multiplying the vector q by the transition probability gives the vector of probabilities at time 1:

$$(0.5 \quad 0.2 \quad 0.3) \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.4 & 0.4 \end{pmatrix} = (0.47 \quad 0.32 \quad 0.21)$$

If $qp = q$ then q is called a stationary distribution. If the distribution at time 0 is the same as the distribution at time 1, then by the Markov property it will be the distribution at all times $n \geq 1$.

Example 17.16. Stationary Distribution for 2- state Markov Chain Recall the weather chain example.

$$\mathbf{W} = \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix}$$

We would like to write it in the stationary distribution form.

$$(\pi_1 \quad \pi_2) \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix} = (\pi_1 \quad \pi_2)$$

This means we get the following system of equations:

$$\begin{aligned} 0.6\pi_1 + 0.2\pi_2 &= \pi_1 \\ 0.4\pi_1 + 0.8\pi_2 &= \pi_2 \end{aligned}$$

From high school algebra, we should be able to solve this as there are 2 equations and 2 unknowns. However, we observe that $\pi_1 = 0.5\pi_2$, so plugging into the second equation we get $\pi_2 = \pi_2$, so this is not the way to solve it. However, we also know that since both equations become $0.4\pi_1 = 0.2\pi_2 \Rightarrow 2\pi_1 = \pi_2$, and $\pi_1 + \pi_2 = 1$, then

$$\pi_1 = \frac{1}{3}, \pi_2 = \frac{2}{3}$$

.

Quickly checking our solution, we arrive at

$$\begin{aligned} (\pi_1 \quad \pi_2) \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix} &= \left(\frac{1}{3} \quad \frac{2}{3}\right) \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix} \\ &= \left(\frac{0.6}{3} + \frac{0.4}{3} \quad \frac{0.4}{3} + \frac{1.6}{3}\right) \\ &= \left(\frac{1}{3} \quad \frac{2}{3}\right) = (\pi_1 \quad \pi_2) \end{aligned}$$

17.2.1 Generalization to Stationary Distribution for 2-state Markov Chains

Now let's try to use this method to a general 2 state Markov chain, where the two states are "1" and "2". Let a be the probability of going from 1 to 2, and b be the probability of going from 2 to 1. Since each row has to sum to 1, this means the general chain looks like

$$\begin{array}{cc} & \begin{array}{cc} \mathbf{1} & \mathbf{2} \end{array} \\ \begin{array}{c} \mathbf{1} \\ \mathbf{2} \end{array} & \begin{array}{cc} 1-a & a \\ b & 1-b \end{array} \end{array}$$

Now to find the stationary distribution, we form the equations

$$(\pi_1 \quad \pi_2) \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix} = (\pi_1 \quad \pi_2)$$

This means we get the following equations:

$$\begin{aligned} (1-a)\pi_1 + b\pi_2 &= \pi_1 \\ a\pi_1 + (1-b)\pi_2 &= \pi_2 \end{aligned}$$

Since both equations simplify to $\pi_1 = \frac{b}{a}\pi_2$, we need to additionally enforce the $\pi_1 + \pi_2 = 1$ condition to solve the systems. Using substitution, we arrive at

$$\frac{b}{a}\pi_2 + \frac{a}{a}\pi_2 = 1 \Rightarrow \pi_2 = \frac{a}{a+b}$$

, so $\pi_1 = \frac{b}{a+b}$. Lastly, we check that this formula is consistent with the answer that we found for the weather chain problem. There $a = 0.4, b = 0.2$, so $\pi_1 = \frac{0.2}{0.4+0.2} = \frac{1}{3}$, and $\pi_2 = \frac{0.4}{0.4+0.2} = \frac{2}{3}$, which is consistent to what we found above.

17.2.2 Stationary Distribution for 3- state Markov Chain

Now we turn our attention to 3-state Markov chains, such as the social mobility problem.

$$\begin{aligned} 0.7\pi_1 + 0.3\pi_2 + 0.2\pi_3 &= \pi_1 \\ 0.2\pi_1 + 0.5\pi_2 + 0.4\pi_3 &= \pi_2 \\ 0.1\pi_1 + 0.2\pi_2 + 0.4\pi_3 &= \pi_3 \end{aligned}$$

But observe that if we add the equations together, since the columns come from a stochastic matrix, we get the redundant equation that $\pi_1 + \pi_2 + \pi_3 = \pi_1 + \pi_2 + \pi_3$, which will cause the system of equations to be unsolvable. So we need to change this to $\pi_1 + \pi_2 + \pi_3 = 1$, since it is a probability distribution, and it must sum to 1. This means that we get the following system of equations.

$$\begin{aligned} -0.3\pi_1 + 0.3\pi_2 + 0.2\pi_3 &= 0 \\ 0.2\pi_1 - 0.5\pi_2 + 0.4\pi_3 &= 0 \\ \pi_1 + \pi_2 + \pi_3 &= 1 \end{aligned}$$